

ME:504

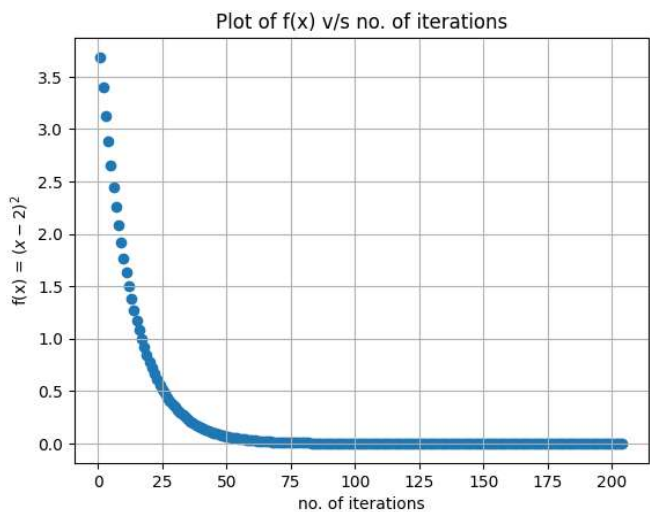
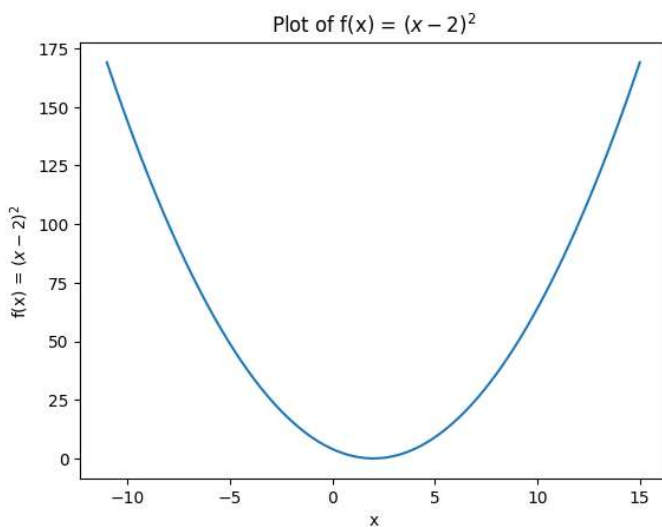
Assignment 1: Result Report

- By default, for all results, unless specified otherwise, Learning Rate = 0.02 has been taken as input by the user.
- Although a similar logic has been implemented for each cell in the kernel, the code has been written separately so that distinct results can be produced by running each cell individually as well.

Question 1

1. $f(x) = (x - 2)^2$

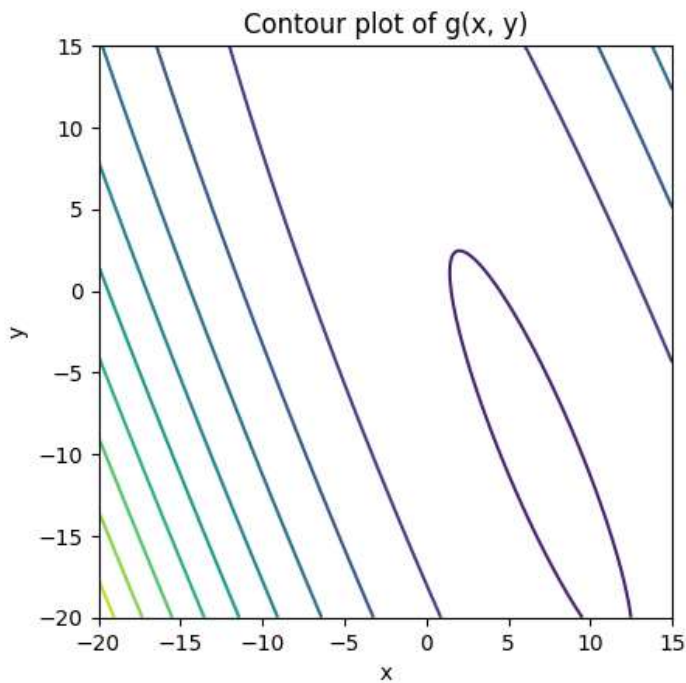
- The point of minima will be reached if we move in the direction opposite to the tangent at the point (i.e, the derivative of the function at that point)



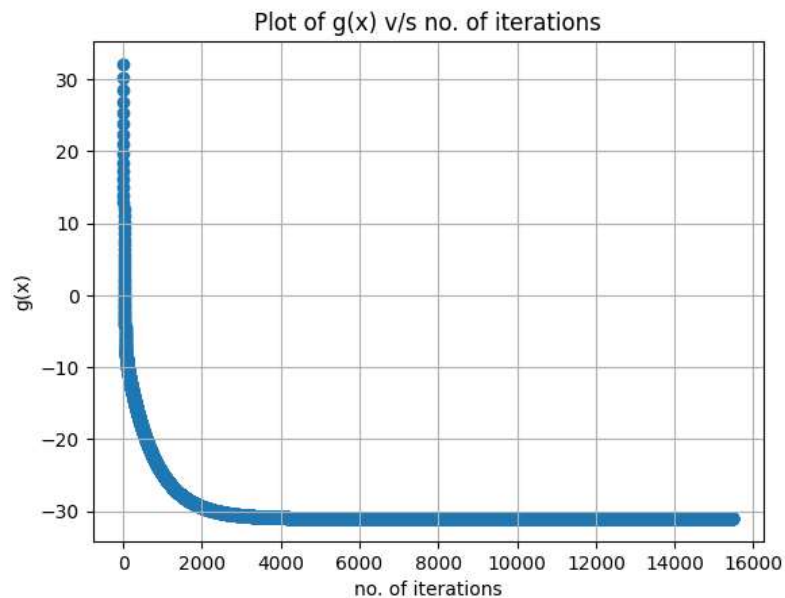
```
The optimum value of x is: 2.0  
The minimum value of function is: 0.0
```

$$2. \ g(x, y) = 5x^2 - 30x + y^2 - 8y + 4xy + 34$$

- The point of minima will be reached if we move in the direction opposite to the normal at the point (i.e, the gradient of the function at that point)



• **Learning rate = 0.002**



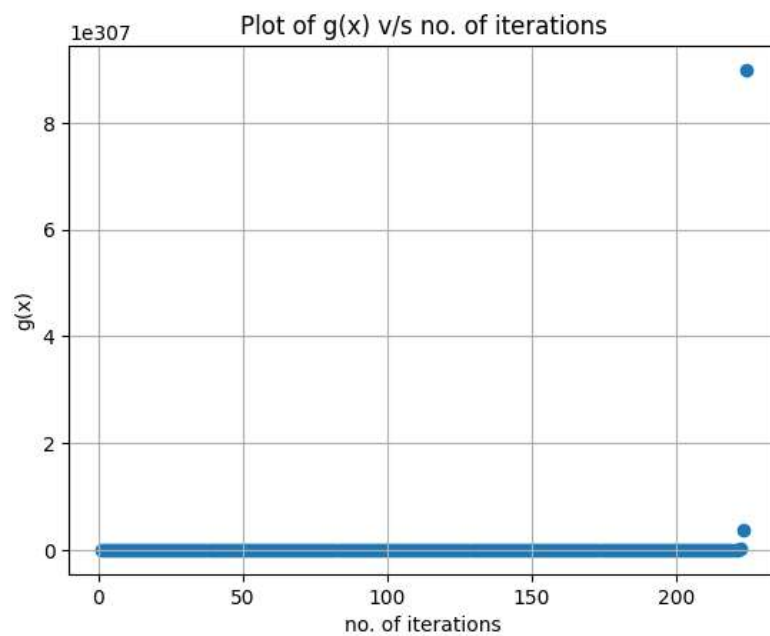
The optimum value of x, y is: 7.0 , -10.0
The minimum value of function is: -31

- Learning rate = 0.02



The optimum value of x, y is: 7.0 , -10.0
The minimum value of function is: -31

- Learning Rate = 0.5

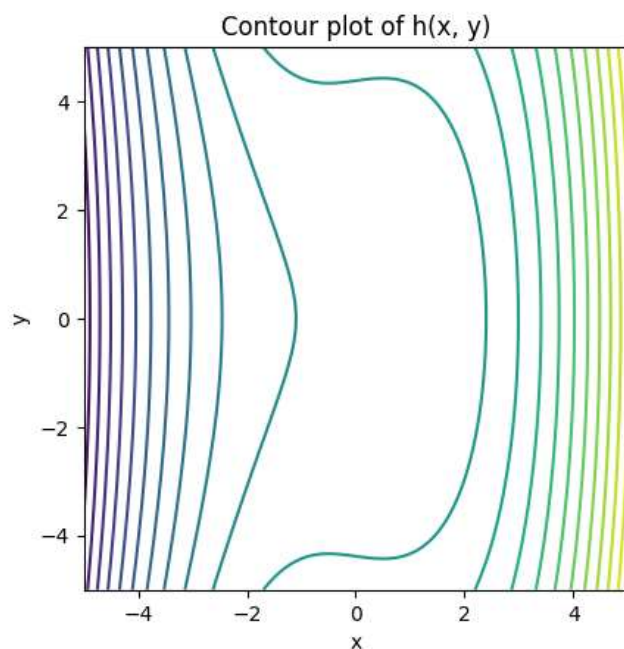


The optimum value of x, y is: nan , inf

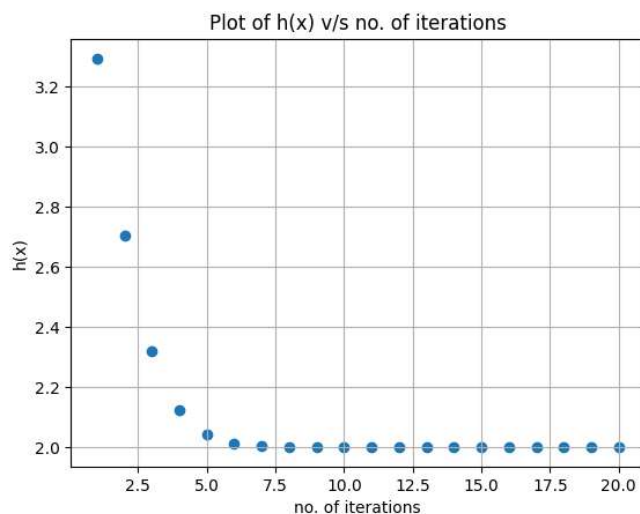
OBSERVATIONS

- For a very small value of learning rate (0.002), a lot of iterations are required to reach optimum value. The graph becomes smoother.
- As the learning rate increases to (0.02), the graph becomes less steep and reaches optimum value in less number of iterations.
- For a very high learning rate (=0.5), error occurs in identifying minima of function. Too many variations in every step leads to a high number of oscillations in the values of objective function with iterations. The code is not able to compile the expression to reach optimum value.

3. $h = 8x^3 + 5y^2 - 6x + 4$



The optimum value of x, y is: 0.5 , 0.0
The minimum value of function is: 2



OBSERVATIONS

- Because this is not a purely convex function, the program finds a local minima as its optimum value.
 - There is no global minima as the function goes to negative infinity.
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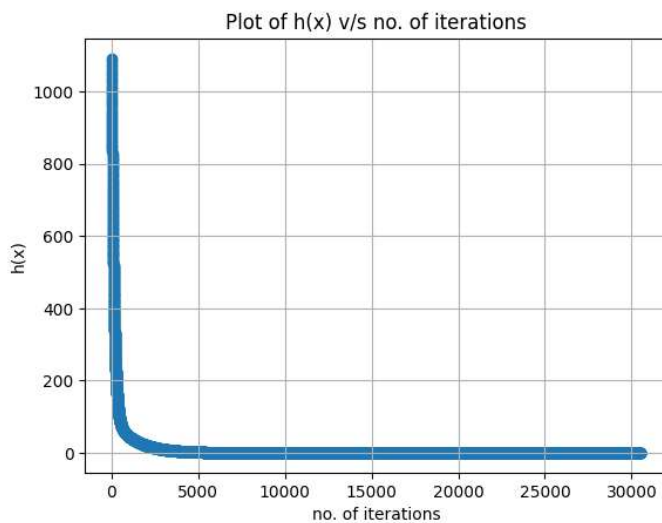
Question 2

Using the function $f(x, y) = (x^2 + ay^2)$

- The values of **a** are 10, 100 and 1000

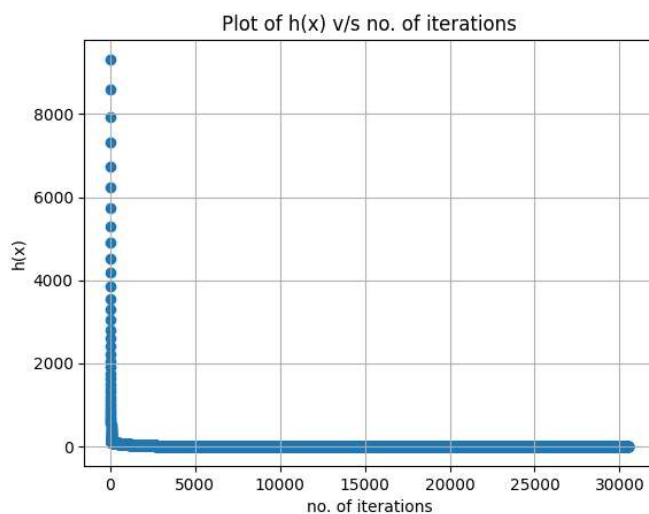
1. Learning Rate is 0.0002

a = 10



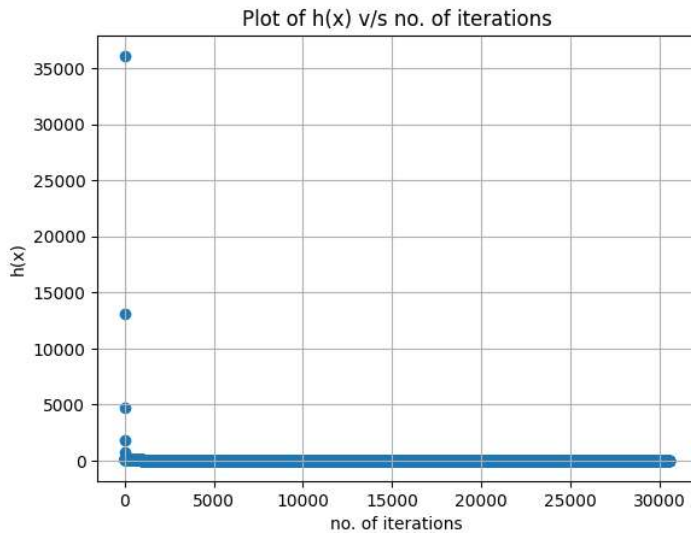
```
Value of a is: 10
The optimum value of x,y is: 0.0 , 0.0
The minimum value of function is: 0
Learning Rate is: 0.0002
```

a = 100



```
Value of a is: 100
The optimum value of x,y is: 0.0 , 0.0
The minimum value of function is: 0
Learning Rate is: 0.0002
```

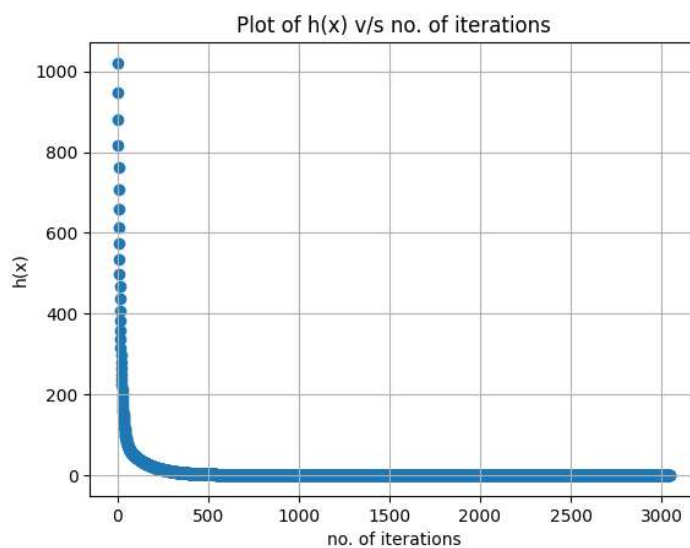
a = 1000



```
Value of a is: 1000
The optimum value of x,y is: 0.0 , 0.0
The minimum value of function is: 0
```

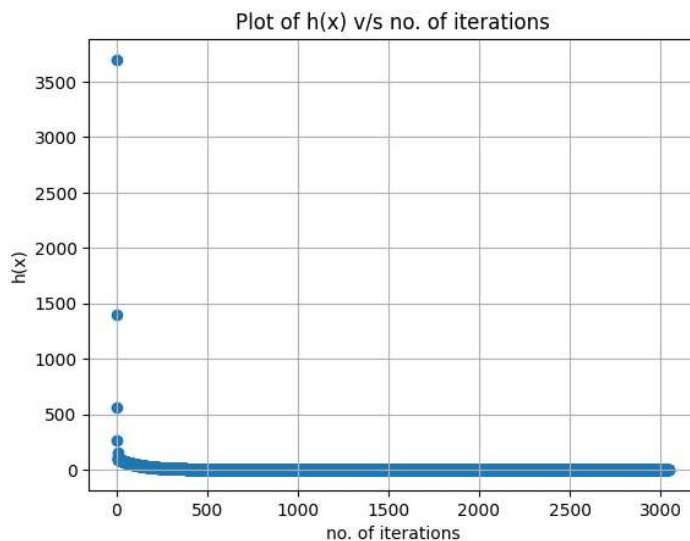
2. Learning Rate is 0.002

a = 10



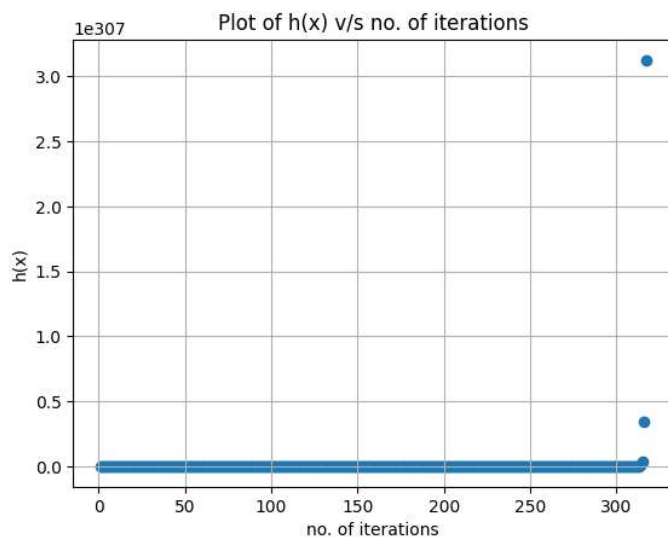
```
Value of a is: 10
The optimum value of x,y is: 0.0 , 0.0
The minimum value of function is: 0
Learning Rate is: 0.002
```

a = 100



```
Value of a is: 100
The optimum value of x,y is: 0.0 , 0.0
The minimum value of function is: 0
Learning Rate is: 0.002
```

a = 1000



```
Value of a is: 1000
The optimum value of x,y is: 0.769 , nan
```

OBSERVATIONS:

As the value of **a** increases,

1. the contour plot of the function becomes more elliptical, with it becoming narrower in the y-direction.
2. the value of gradient of function with respect to y also increases. Thus gradient descent oscillates more for a fixed learning rate, leading to more decrease in objective function with iterations.
3. a smaller value of learning rate is preferred to perform a smoother gradient descent algorithm on the function, and to avoid error in final result because of over - oscillations in the program.

